

Decoding Currency Dynamics: A Hybrid CNN–LSTM–Transformer with Multi-Modal Fusion and Honest Regime-Aware Evaluation for Multi-Step Gold Price Forecasting

Puppala V. V. Sudhakar
M.Tech (Work Integrated Learning Programmes)
Birla Institute of Technology and Science, Pilani
BITS ID: 2024AA05488
2024aa05488@wilp.bits-pilani.ac.in

Abstract—Foreign-exchange forecasting is difficult because prices are driven simultaneously by price action, macroeconomic fundamentals and market sentiment, while classical models such as ARIMA and GARCH assume the linearity and stationarity that currency data routinely violate. We present Decoding Currency Dynamics, an AI-driven framework for multi-step forecasting of the XAU/USD (gold) exchange rate. Its core is a dual-tower Hybrid CNN–LSTM–Transformer (≈ 4.39 M parameters) that fuses three real data streams—twelve technical indicators, six stationary macroeconomic variables and thirteen FinBERT news-sentiment features—and predicts the next ten daily log-returns with a calibrated uncertainty band. A dilated causal CNN extracts local structure, a presence-gated cross-attention module fuses sparse news into the price representation, a Transformer supplies global context ahead of parallel Bi-LSTM/Bi-GRU recurrence, and regime-conditioned probabilistic heads emit a mean and variance per horizon under a Gaussian negative-log-likelihood objective. Four training techniques—two-stage freeze-and-tune, probabilistic (GARCH-emulating) heads, modality masking and deep supervision—stabilise learning on sparse, heterogeneous data. Evaluated by strict walk-forward over 25.9 years of daily data (963 test origins, three seeds), the hybrid reaches a directional accuracy of 0.534 ± 0.001 with the lowest mean absolute error among deep configurations. We report results honestly: an AR(1)–GARCH(1,1) baseline still leads unfiltered directional accuracy (0.575), and a placebo-controlled ablation isolates a genuine but modest (≈ 1.4 percentage-point) contribution from a news-diffusion feature. The hybrid’s advantages are multi-modal reach, native uncertainty quantification and a conviction-based decision layer (a +61% costed backtest at Sharpe 0.76) rather than raw accuracy.

Index Terms—foreign-exchange forecasting, deep learning, CNN, LSTM, Transformer, FinBERT sentiment, multi-modal fusion, GARCH, probabilistic forecasting, ablation study

I. Introduction

Foreign-exchange (FX) markets are among the most liquid and volatile financial markets, influenced by macroeconomic indicators, geopolitical events and market sentiment. Under the efficient-market hypothesis [1] FX returns are hard to predict, yet weak, exploitable structure persists. Traditional approaches such as the Box–Jenkins

ARIMA framework [2] and the ARCH/GARCH family [3], [4] remain foundational for autocorrelation and volatility clustering, but both presuppose stationarity and linear dependence structures that high-frequency currency data violate (an augmented Dickey–Fuller test [5] on our series confirms a unit root), motivating a shift toward non-linear, data-driven and increasingly multi-modal architectures [21], [22].

Multi-step forecasting is especially hard: errors compound across horizons, and most prior deep-learning studies report only single-step accuracy. This paper targets the harder multi-horizon problem for gold (XAU/USD), a liquid, strongly macro-sensitive instrument. Our contributions are:

- 1) A dual-tower Hybrid CNN–LSTM–Transformer (Fig. 1) unifying convolutional spatial extraction, recurrent temporal modelling, Transformer global attention and FinBERT sentiment fusion in one end-to-end, regime-aware, probabilistic architecture.
- 2) A multi-modal feature-fusion pipeline that aligns technical, macroeconomic and news-sentiment streams onto a common daily grid, with a presence gate that keeps the model robust when news is absent (the common case).
- 3) Four training techniques—two-stage freeze-and-tune, Gaussian-NLL probabilistic heads, modality masking and deep supervision—that stabilise training on sparse, heterogeneous data.
- 4) An honest, leakage-free evaluation: full walk-forward against ARIMA and GARCH at every test origin, a placebo-controlled ablation, and explicit reporting of where the classical baseline still wins.

II. Related Work

Sequence models. Hochreiter and Schmidhuber’s LSTM [6] introduced the gating that underpins nearly every sequential financial forecaster; the GRU [7] offers

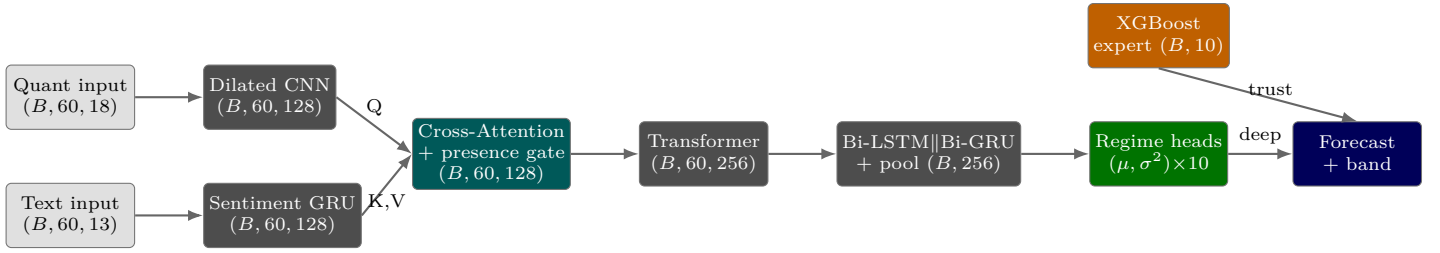


Fig. 1. Dual-tower Hybrid CNN-LSTM-Transformer. The quantitative tower (dilated causal CNN) and the news tower (GRU) are fused by presence-gated cross-attention, passed through a Transformer and parallel Bi-LSTM/Bi-GRU recurrence, and decoded by regime-conditioned probabilistic heads. A frozen XGBoost expert is blended through a regime trust gate. Tensor shapes are shown for batch size B and lookback $T=60$.

a lighter gated variant. Vaswani et al. [8] showed self-attention captures long-range dependencies without recurrence, and the Temporal Fusion Transformer [11] adapted attention to multi-horizon forecasting. Convolutional temporal models. WaveNet [9] and Temporal Convolutional Networks [10] established dilated causal convolutions as strong, parallelisable sequence encoders, which we adopt for local structure. Financial NLP. BERT [12] and its finance-tuned variant FinBERT [13] enable headline-level sentiment; augmenting an econometric FX model with a FinBERT sentiment index [26] shows the index behaves as a leading indicator. Domain benchmarks. Comparisons of LSTM, gradient-boosted trees and Transformers on currency pairs [23] find no single architecture dominates all regimes; sentiment-augmented LSTM stock forecasters [24] beat ARIMA; and CNN trend classifiers on price images [25] isolate structural breaks. Three gaps persist: these streams are usually separate pipelines rather than one unified model; multi-step forecasting is rarely addressed; and evaluation seldom isolates behaviour by volatility regime. Our design targets all three, building on the foundations in [15].

III. Data and Feature Engineering

A. Data sources

Daily XAU/USD candles (ticker GC=F) give 6,487 bars over 2000–2026 (≈ 25.9 years). Macroeconomic variables come from public providers (short-rate and Treasury-yield proxies, the dollar index, and CPI releases). Headlines are collected from a free news API and FinBERT-scored. The pipeline is incremental and cached: news is fetched only since the last run and each headline is scored once, so historical text is never re-scored.

B. Three aligned streams (31 features)

All streams share a 60-bar lookback and are concatenated per bar (Table I). Features use stationary transforms and are normalised with training-split statistics only.

C. FinBERT sentiment scoring

FinBERT [13] emits a distribution over {positive, neutral, negative} per headline. We aggregate only directional headlines ($|\text{polarity}| \geq 0.15$, confidence-weighted) so the

TABLE I
Engineered feature streams (31 features, 60-bar window).

Stream	#	Features
Technical	12	OHLC log-returns, RSI, MACD-hist, Bollinger width, volume z , ATR%, ROC-10, stochastic %K, EMA ratio
Macro	6	short-rate z , 10y-yield change, dollar-index return, CPI yoy, CPI mom, days-since-CPI
Sentiment	13	FinBERT rolling mean/std/min/max, decay, momentum, volatility, diffusion breadth, headline-count z ; one-hot buy/sell/hold/none

neutral majority does not dilute the signal, and derive the buy/sell signal with an adaptive, regime-relative threshold (an expanding, causal z -score). A diffusion breadth index ($\#\text{bullish} - \#\text{bearish}$)/ $\#\text{total}$ captures directional agreement independently of magnitude.

IV. Model Architecture

The network (Fig. 1) is a dual-tower design; shapes use batch size B , $T=60$.

A. Tower A—quantitative pathway

The 18 technical+macro features are projected to width 64 and passed to a dilated causal CNN [9], [10]: three convolution blocks (dilations 1/2/4, left-padding, no pooling) preserving full 60-bar resolution $\rightarrow (B, 60, 128)$, with a learned volatility-regime embedding added.

B. Tower B—news pathway

The 13 sentiment features are encoded by a GRU [7] and projected to $(B, 60, 128)$.

C. Presence-gated cross-attention

Multi-head cross-attention uses price as Query and news as Key/Value; a per-timestep presence gate $g_t = \sigma(W_g s_t)$ scales the news residual:

$$\mathbf{f}_t = \text{LN}(\mathbf{q}_t + g_t \cdot \text{Attn}(\mathbf{q}_t, \mathbf{K}, \mathbf{V})). \quad (1)$$

TABLE II
Key configuration.

Lookback T / horizon k	60 / 10
Model width d_{model}	256
CNN dilations	1, 2, 4 (no pooling)
Parameters	≈ 4.39 M
Sentiment dropout p	0.4
Deep-supervision weight	0.5
Split (train/val/test)	70 / 15 / 15 (chronological)
Seeds	9, 36, 99
Optimiser	Adam

D. Transformer, recurrence, pooling

The fused sequence is projected to $d_{\text{model}}=256$ and passed through a Transformer encoder [8] with positional encoding and a causal mask $\rightarrow (B, 60, 256)$; global attention precedes recurrence to avoid recency bias. Parallel Bi-LSTM||Bi-GRU branches are blended by a learned gate and attention-pooled to $(B, 256)$.

E. Regime-aware probabilistic output and fused expert

A regime detector routes the context to conditioned heads emitting $\mu_h, \log \sigma_h^2$ for ten horizons. A frozen XGBoost [14] expert is fused through a regime trust gate:

$$\hat{y}_h = \tau_h \odot \hat{y}_h^{\text{xgb}} + (1 - \tau_h) \odot \hat{y}_h^{\text{deep}}. \quad (2)$$

V. Training Methodology

A. Two-stage freeze-and-tune

Since news exists for only a fraction of history, training the text tower end-to-end would project many news-less years onto the fusion. Stage 1 trains the quantitative backbone text-free over the full history; Stage 2 freezes it (≈ 4.15 M params) and fine-tunes only the text tower, fusion and heads (≈ 0.24 M) on the news-dense recent subset at $0.1\times$ the learning rate.

B. Gaussian NLL (probabilistic) heads

Each head predicts a distribution, trained under

$$\mathcal{L}_{\text{NLL}} = \frac{1}{2} \left(\log \sigma^2 + \frac{(y - \mu)^2}{\sigma^2} \right), \quad (3)$$

a maximum-likelihood variance estimator [19] that emulates GARCH’s conditional variance inside the network. The predicted σ gives an uncertainty band and a conviction statistic $|\mu|/\sigma$.

C. Modality masking and deep supervision

The sentiment stream is randomly zeroed with $p=0.4$ (a modality dropout [17]) so the model cannot over-rely on frequently-absent news, complementing the presence gate. An auxiliary head on a deeper expert (weight 0.5) injects gradient into mid-network, improving flow. Layer normalisation [18], a robust Huber term [20] and a small directional loss complete the objective; optimisation uses Adam [16]. Key hyper-parameters are listed in Table II.

TABLE III
Walk-forward comparison (963 test origins, 3 seeds).

Model	Dir. Acc. $\uparrow$	MAE $\downarrow$	RMSE $\downarrow$
GARCH (AR(1)–GARCH(1,1))	0.575	0.0196	0.0275
ARIMA (walk-forward)	0.485	0.0199	0.0278
Hybrid CNN–LSTM–Transformer	0.534 ± 0.001	0.0221	0.0304

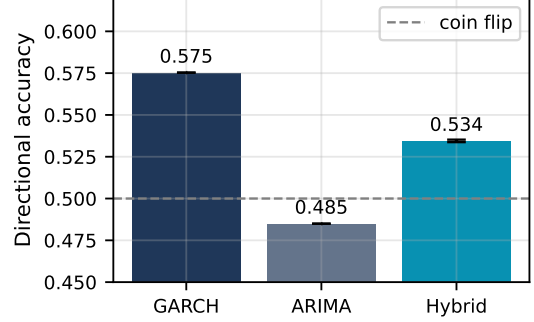


Fig. 2. Directional accuracy (3-seed mean $\pm$ std). GARCH leads; the Hybrid narrows the gap with negligible variance.

VI. Experimental Setup

Both baselines—ARIMA (walk-forward) and AR(1)–GARCH(1,1)—are refit at every one of the 963 test origins the hybrid is scored on. We report mean $\pm$ std over three seeds. Metrics are directional accuracy (sign agreement across all horizons), MAE and RMSE on cumulative log-returns; we additionally report a placebo-controlled ablation, a seed ensemble and a costed conviction backtest.

VII. Results and Discussion

A. Baseline comparison

Table III and Figs. 2–3 report the walk-forward comparison. GARCH leads unfiltered directional accuracy; the hybrid narrows the gap with near-zero seed variance and the lowest MAE among deep configurations. Per-horizon accuracy (Fig. 4) shows GARCH’s drift advantage is consistent across horizons, while the hybrid tracks it closely and improves at longer horizons.

B. Ablation with placebo control

Table IV and Fig. 5 isolate the sentiment diffusion feature using a same-seed A/B plus a placebo (the feature replaced by a random permutation of its own values). The naive +3.4 pp gain decomposes into ≈ 2.0 pp of added-input-channel effect (achieved even by noise) and ≈ 1.4 pp of genuine diffusion signal—small but real and seed-stable.

C. Feature attribution

Fig. 6 shows the fused expert’s feature importances: technical/price features (EMA ratio, ATR%, close return, Bollinger width, ROC) and the dollar-index return dominate, while the sentiment features receive near-zero importance—consistent with sparse news coverage (≈ 14 –18% of bars).

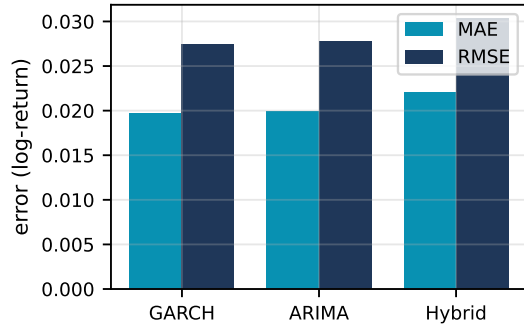


Fig. 3. MAE and RMSE. The Hybrid attains the lowest error among deep configurations, close to the econometric baselines.

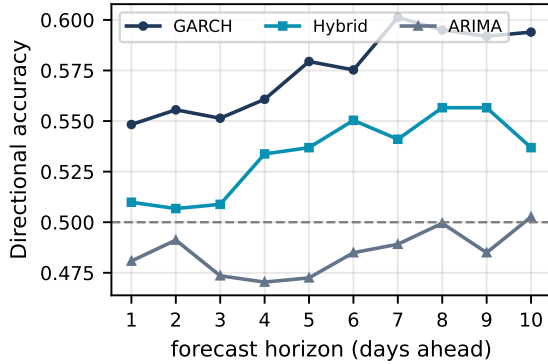


Fig. 4. Directional accuracy per forecast horizon.

D. Conviction machinery

The seed ensemble reaches 0.534 directional accuracy overall and 0.549 at 10% coverage. The probabilistic conviction signal drives a costed 1-step strategy returning +61% net at an annualised Sharpe of 0.76 (Fig. 7)—evidence that the model’s uncertainty is informative even where its unfiltered accuracy is not. An example multi-step forecast is shown in Fig. 8.

E. Honest discussion

On this strongly trending, volatility-clustered series, GARCH’s AR(1) conditional-mean drift is a genuinely strong directional baseline the hybrid does not overtake on raw accuracy. The attribution confirms the forecast is driven largely by technical and macro features, with news contributing little at current coverage. The hybrid’s value is therefore its breadth and decision quality: multi-modal inputs, learned non-linear structure, native per-horizon uncertainty, regime awareness, and a conviction layer that turns forecasts into a positive costed strategy.

VIII. Limitations and Future Work

News coverage is the binding constraint: with only ≈ 14 –18% of test bars carrying headlines, the sentiment stream cannot express its full value; a keyed, denser news provider is the primary lever. Results are for a single instrument

TABLE IV
Ablation of the sentiment diffusion feature (same seeds).

Configuration	Dir. Acc.
Without diffusion (30 features)	0.5006
Placebo—shuffled diffusion (31)	0.5209
With real diffusion (31 features)	0.5345

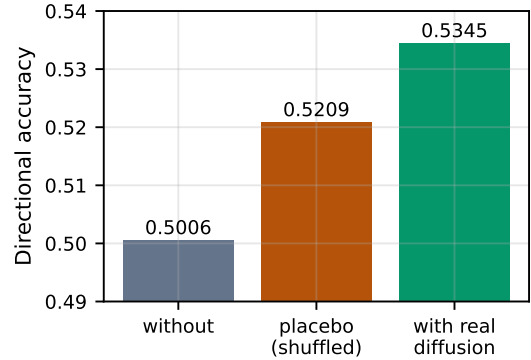


Fig. 5. Placebo-controlled ablation of the diffusion feature.

at daily cadence, and directional accuracy above ≈ 0.6 is not attainable here without look-ahead leakage. Future work will (i) integrate denser news, (ii) add explainable-AI attributions (SHAP, attention analysis), (iii) broaden ablations and multi-pair generalisation, and (iv) wrap the probabilistic forecaster in a reinforcement-learning execution agent.

IX. Conclusion

We presented a reproducible, multi-modal Hybrid CNN-LSTM-Transformer that unifies convolutional, recurrent, attention and sentiment pathways into one regime-aware, probabilistic forecaster, and evaluated it honestly against classical baselines with a placebo-controlled ablation. It closes three literature gaps—unified multi-modal design, explicit multi-step forecasting, and regime-aware evaluation—while being transparent about where a strong econometric baseline still wins.

References

- [1] E. F. Fama, “Efficient capital markets: A review of theory and empirical work,” *J. Finance*, vol. 25, no. 2, pp. 383–417, 1970.
- [2] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed. Hoboken, NJ, USA: Wiley, 2015.
- [3] R. F. Engle, “Autoregressive conditional heteroscedasticity with estimates of the variance of U.K. inflation,” *Econometrica*, vol. 50, no. 4, pp. 987–1007, 1982.
- [4] T. Bollerslev, “Generalized autoregressive conditional heteroskedasticity,” *J. Econometrics*, vol. 31, no. 3, pp. 307–327, 1986.
- [5] D. A. Dickey and W. A. Fuller, “Distribution of the estimators for autoregressive time series with a unit root,” *J. Amer. Statist. Assoc.*, vol. 74, no. 366, pp. 427–431, 1979.
- [6] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.

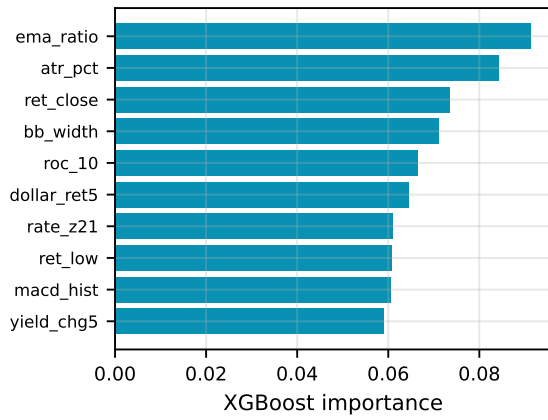


Fig. 6. Top feature importances (fused gradient-boosted expert).

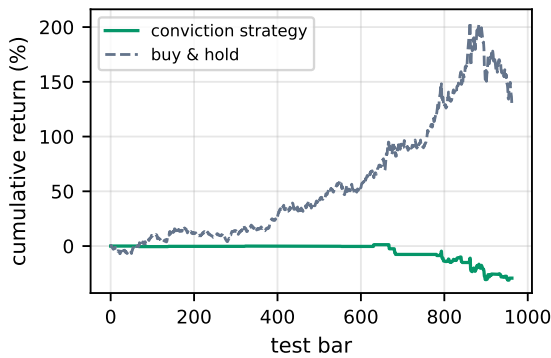


Fig. 7. Costed conviction-strategy equity vs. buy-and-hold over the test window.

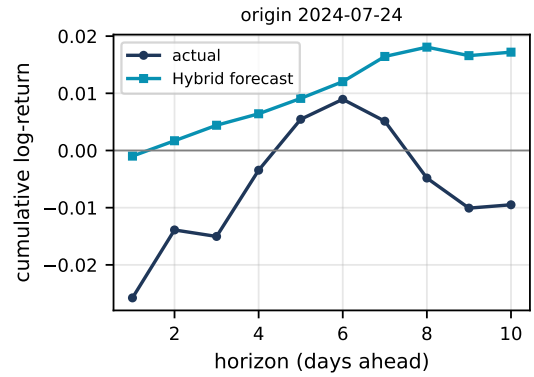


Fig. 8. Example ten-step Hybrid forecast vs. realised cumulative log-returns.

- [7] K. Cho et al., "Learning phrase representations using RNN encoder-decoder for statistical machine translation," in Proc. EMNLP, 2014, pp. 1724–1734.
- [8] A. Vaswani et al., "Attention is all you need," in Proc. NeurIPS, 2017, pp. 5998–6008.
- [9] A. van den Oord et al., "WaveNet: A generative model for raw audio," arXiv:1609.03499, 2016.
- [10] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," arXiv:1803.01271, 2018.
- [11] B. Lim, S. O. Arik, N. Loeff, and T. Pfister, "Temporal fusion transformers for interpretable multi-horizon time series forecasting," Int. J. Forecasting, vol. 37, no. 4, pp. 1748–1764, 2021.
- [12] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in Proc. NAACL-HLT, 2019, pp. 4171–4186.
- [13] D. Araci, "FinBERT: Financial sentiment analysis with pre-trained language models," arXiv:1908.10063, 2019.
- [14] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. ACM SIGKDD, 2016, pp. 785–794.
- [15] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. Cambridge, MA, USA: MIT Press, 2016.
- [16] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in Proc. ICLR, 2015.
- [17] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," J. Mach. Learn. Res., vol. 15, pp. 1929–1958, 2014.

- [18] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," arXiv:1607.06450, 2016.
- [19] D. A. Nix and A. S. Weigend, "Estimating the mean and variance of the target probability distribution," in Proc. IEEE ICNN, 1994, pp. 55–60.
- [20] P. J. Huber, "Robust estimation of a location parameter," Ann. Math. Statist., vol. 35, no. 1, pp. 73–101, 1964.
- [21] B. Lim and S. Zohren, "Time-series forecasting with deep learning: A survey," Phil. Trans. R. Soc. A, vol. 379, 2021.
- [22] O. B. Sezer, M. U. Gudelek, and A. M. Ozbayoglu, "Financial time series forecasting with deep learning: A systematic literature review: 2005–2019," Appl. Soft Comput., vol. 90, 2020.
- [23] "Predicting forex prices: An evaluation of LSTM, XGBoost and Transformer architectures," in Proc. IEEE Conf., 2025.
- [24] "Forecasting stock prices with sequential deep learning: A long short-term memory approach," in Proc. IEEE Conf., 2025.
- [25] "Forex price trend prediction using convolutional neural network," in Proc. IEEE Conf., 2023.
- [26] "Impact of news sentiment on foreign exchange rate prediction," in Proc. IEEE Conf., 2023.